Riemann–Roch on the Projective Line

A Book-Style Introduction with 50 Exercises

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1. Introduction

The purpose of this chapter is to understand the Riemann–Roch theorem in the simplest compact Riemann surface,

$$X = \mathbb{CP}^1.$$

For an arbitrary compact Riemann surface X of genus g, the Riemann–Roch theorem states that

$$\ell(D) - \ell(K - D) = \deg D + 1 - g,$$

where D is a divisor on X, K is a canonical divisor, and

$$L(D) = \{f \in \mathcal{M}(X) \setminus \{0\} : (f) + D \geq 0\} \cup \{0\}, \qquad \ell(D) = \dim_{\mathbb{C}} L(D).$$

On the projective line the genus is 0, so the formula becomes

$$\ell(D) - \ell(K - D) = \deg D + 1.$$

What makes $\mathbb{CP}^1$ special is that every relevant object can be computed explicitly. Meromorphic functions are rational functions, every divisor class is determined by degree, the canonical divisor is easy to write down, and the spaces $L(D)$ can be described by elementary polynomial calculations. Consequently, the theorem becomes a concrete identity rather than a mysterious abstract statement.

The chapter proceeds as follows. We begin with the geometry of $\mathbb{CP}^1$, then study meromorphic functions and divisors, compute divisor classes and spaces of sections, determine the canonical divisor, and finally prove Riemann–Roch in this setting. The last section contains fifty exercises arranged in increasing order of difficulty.

2. The projective line

2.1. Definition and affine charts

Definition 2.1. The *complex projective line* is the quotient

$$\mathbb{CP}^1 = (\mathbb{C}^2 \setminus \{(0,0)\})/\mathbb{C}^\times,$$

where $\lambda \in \mathbb{C}^\times$ acts by

$$\lambda \cdot (Z_0, Z_1) = (\lambda Z_0, \lambda Z_1).$$

The class of (Z_0, Z_1) is denoted by $[Z_0 : Z_1]$.

The two standard affine charts are

$$U_0 = \{[Z_0 : Z_1] \in \mathbb{CP}^1 : Z_0 \neq 0\}, \qquad U_\infty = \{[Z_0 : Z_1] \in \mathbb{CP}^1 : Z_1 \neq 0\}.$$

On U_0 one uses the coordinate

$$z = \frac{Z_1}{Z_0},$$

and on U_∞ one uses

$$w = \frac{Z_0}{Z_1}.$$

On the overlap $U_0 \cap U_\infty$ these coordinates satisfy

$$w = \frac{1}{z}.$$

Thus $\mathbb{CP}^1$ is obtained by gluing two copies of $\mathbb{C}$ by inversion.

Remark 2.2. The distinguished point

$$\infty = [0 : 1]$$

lies in U_∞ and corresponds to $w = 0$. In the chart $U_0 \cong \mathbb{C}$, the projective line looks like the complex plane together with one additional point at infinity.

2.2. Meromorphic functions on $\mathbb{CP}^1$

Proposition 2.3. *Every meromorphic function on $\mathbb{CP}^1$ is a rational function in the affine coordinate z. Equivalently,*

$$\mathcal{M}(\mathbb{CP}^1) = \mathbb{C}(z).$$

Proof. Every rational function in z defines a meromorphic function on $U_0 \cong \mathbb{C}$, and after the substitution $z = 1/w$ it is meromorphic near $w = 0$, hence on all of $\mathbb{CP}^1$.

Conversely, let $f \in \mathcal{M}(\mathbb{CP}^1)$. Restricting to the chart $U_0 \cong \mathbb{C}$, we obtain a meromorphic function on $\mathbb{C}$. Since the only point omitted by the affine chart is ∞, the function has at worst an isolated singularity there. A meromorphic function on $\mathbb{C}$ with at worst a pole at infinity is rational. Therefore $f \in \mathbb{C}(z)$. □

Remark 2.4. This is the first great simplification of the genus-zero case: all global meromorphic data are encoded by elementary algebra in one variable.

3. Divisors on the projective line

3.1. Orders and principal divisors

Definition 3.1. A *divisor* on $\mathbb{CP}^1$ is a finite formal sum

$$D = \sum_{p \in \mathbb{CP}^1} n_p [p], \qquad n_p \in \mathbb{Z}.$$

Its *degree* is

$$\deg D = \sum_{p \in \mathbb{CP}^1} n_p.$$

If $f \in \mathcal{M}(\mathbb{CP}^1)^\times$, then its *principal divisor* is

$$(f) = \sum_{p \in \mathbb{CP}^1} \operatorname{ord}_p(f)[p].$$

Example 3.2. Consider the function $z - a$ on U_0. It has a simple zero at $a \in \mathbb{C}$. To compute its order at ∞, write $z = 1/w$. Then

$$z - a = \frac{1 - aw}{w}.$$

Near $w = 0$ the factor $1 - aw$ is a unit and $1/w$ has a simple pole. Hence

$$(z - a) = [a] - [\infty].$$

Example 3.3. For distinct $a, b \in \mathbb{C}$,

$$\left(\frac{z-a}{z-b}\right) = [a] - [b].$$

This is an elementary but fundamental example of a principal divisor.

Lemma 3.4. *If $m \in \mathbb{Z}$, then*

$$(z^m) = m[0] - m[\infty].$$

Proof. For $m \geq 0$, the function z^m has a zero of order m at 0 and, since $z = 1/w$ near ∞, a pole of order m at ∞. The case $m < 0$ follows by applying the positive case to z^{-m} and then negating the divisor. □

Proposition 3.5. *Every principal divisor on $\mathbb{CP}^1$ has degree 0.*

Proof. Let $f = P/Q$ with coprime polynomials $P, Q \in \mathbb{C}[z]$, where $\deg P = m$ and $\deg Q = n$. The total order of the finite zeros of f is m, and the total order of the finite poles is $-n$. Near ∞ we have

$$f(1/w) = \frac{P(1/w)}{Q(1/w)} \sim c\, w^{n-m},$$

so

$$\operatorname{ord}_\infty(f) = n - m.$$

Hence

$$\deg(f) = m - n + (n - m) = 0.$$

□

3.2. Linear equivalence

Definition 3.6. Two divisors D, E are *linearly equivalent*, written $D \sim E$, if there exists $f \in \mathcal{M}(\mathbb{CP}^1)^\times$ such that

$$D - E = (f).$$

Theorem 3.7. *Every divisor on $\mathbb{CP}^1$ is linearly equivalent to a unique multiple of $[\infty]$. More precisely, for every divisor D,*

$$D \sim (\deg D)[\infty].$$

Consequently,

$$\operatorname{Pic}(\mathbb{CP}^1) \cong \mathbb{Z}, \qquad [D] \longmapsto \deg D.$$

Proof. Write the divisor in the form

$$D = \sum_{i=1}^{r} m_i[a_i] - \sum_{j=1}^{s} n_j[b_j] + c[\infty],$$

where $a_i, b_j \in \mathbb{C}$ and $m_i, n_j > 0$. Consider the rational function

$$h(z) = \frac{\prod_{i=1}^{r}(z-a_i)^{m_i}}{\prod_{j=1}^{s}(z-b_j)^{n_j}}.$$

Using $(z-a) = [a] - [\infty]$, we obtain

$$(h) = \sum_{i=1}^{r} m_i[a_i] - \sum_{j=1}^{s} n_j[b_j] - \left(\sum_{i=1}^{r} m_i - \sum_{j=1}^{s} n_j\right)[\infty].$$

Subtracting, we find

$$D - (h) = \left(c + \sum_{i=1}^{r} m_i - \sum_{j=1}^{s} n_j\right)[\infty] = (\deg D)[\infty].$$

Thus $D \sim (\deg D)[\infty]$.

Uniqueness follows from degree: if $m[\infty] \sim n[\infty]$, then $(m-n)[\infty]$ is principal and therefore has degree 0, so $m = n$. □

Remark 3.8. This theorem is the divisor-theoretic incarnation of the statement that line bundles on $\mathbb{CP}^1$ are classified by degree.

4. The spaces $L(D)$

4.1. Definition and first reduction

Definition 4.1. For a divisor D on $\mathbb{CP}^1$, define

$$L(D) = \{f \in \mathcal{M}(\mathbb{CP}^1)^\times : (f) + D \geq 0\} \cup \{0\}.$$

Its dimension over $\mathbb{C}$ is denoted by

$$\ell(D) = \dim_{\mathbb{C}} L(D).$$

The meaning of $L(D)$ is straightforward: it is the space of meromorphic functions whose poles are bounded by D. Since every divisor is linearly equivalent to $n[\infty]$, the study of $L(D)$ reduces to the study of the special divisors $n[\infty]$.

Lemma 4.2. *If $D \sim E$, then multiplication by a suitable meromorphic function induces an isomorphism*

$$L(D) \cong L(E).$$

In particular, $\ell(D) = \ell(E)$.

Proof. If $D - E = (h)$, define

$$T_h \colon L(E) \longrightarrow L(D), \qquad f \longmapsto fh.$$

Then

$$(f) + E \geq 0 \iff (fh) + D = (f) + (h) + D = (f) + E \geq 0,$$

so T_h is well-defined. Its inverse is multiplication by h^{-1}. □

4.2. Explicit computation of $L(n[\infty])$

Proposition 4.3. *For $n \in \mathbb{Z}$,*

$$L(n[\infty]) = \begin{cases} 0, & n < 0, \\ \{\textit{polynomials in } z \textit{ of degree at most } n\}, & n \geq 0. \end{cases}$$

Consequently,

$$\ell(n[\infty]) = \begin{cases} 0, & n < 0, \\ n+1, & n \geq 0. \end{cases}$$

Proof. Assume first that $n \geq 0$. Let $f \in L(n[\infty])$. Then f has no poles on $\mathbb{C}$, so it is an entire function in the coordinate z. The condition $(f) + n[\infty] \geq 0$ says that the pole order of f at ∞ is at most n. Therefore f is a polynomial of degree at most n.

Conversely, every polynomial of degree at most n has pole order at most n at ∞, so it belongs to $L(n[\infty])$.

If $n < 0$, then every element of $L(n[\infty])$ would have to vanish at ∞ and be holomorphic on all of $\mathbb{CP}^1$. But every global holomorphic function on a compact Riemann surface is constant, and no nonzero constant can vanish at ∞. Hence $L(n[\infty]) = 0$. □

Corollary 4.4. *If D is any divisor on $\mathbb{CP}^1$ and $d = \deg D$, then*

$$\ell(D) = \begin{cases} 0, & d < 0, \\ d+1, & d \geq 0. \end{cases}$$

Proof. Since $D \sim d[\infty]$, the preceding lemma gives $\ell(D) = \ell(d[\infty])$, and the result follows from the proposition. □

5. The line bundles $\mathcal{O}(n)$

5.1. Transition functions

The divisor $n[\infty]$ corresponds to a line bundle usually denoted by $\mathcal{O}(n)$. In concrete terms, it is described on the standard cover $\{U_0, U_\infty\}$ by the transition function z^n on the overlap.

A holomorphic section of $\mathcal{O}(n)$ is given by a pair of holomorphic functions

$$s_0 \in \mathcal{O}(U_0), \qquad s_\infty \in \mathcal{O}(U_\infty),$$

satisfying

$$s_\infty(w) = w^n s_0(1/w) \qquad \text{on } U_0 \cap U_\infty.$$

Proposition 5.1. *For every integer n,*

$$H^0(\mathbb{CP}^1, \mathcal{O}(n)) \cong L(n[\infty]).$$

In particular,

$$h^0(\mathbb{CP}^1, \mathcal{O}(n)) = \begin{cases} 0, & n < 0, \\ n+1, & n \geq 0. \end{cases}$$

Proof. A meromorphic function $f \in L(n[\infty])$ may be interpreted as a meromorphic section of the bundle associated with the divisor $n[\infty]$, and the condition $(f) + n[\infty] \geq 0$ says exactly that this section is holomorphic. Conversely, every holomorphic section determines such a meromorphic function.

More concretely, if $n \geq 0$, then $s_0(z)$ must be a polynomial of degree at most n, because the expression $w^n s_0(1/w)$ has to extend holomorphically across $w = 0$. Thus the space of sections has basis

$$1, z, z^2, \ldots, z^n.$$

If $n < 0$, no nonzero holomorphic section can satisfy the gluing condition. □

Remark 5.2. The bundles $\mathcal{O}(n)$ are the basic line bundles on the projective line. Every holomorphic line bundle on $\mathbb{CP}^1$ is isomorphic to exactly one of them.

6. The canonical divisor

6.1. Meromorphic differentials

A meromorphic 1-form on $\mathbb{CP}^1$ can be written in the affine chart as

$$\omega = f(z)\, dz, \qquad f(z) \in \mathbb{C}(z).$$

To compute the divisor of dz, write $z = 1/w$. Then

$$dz = d\left(\frac{1}{w}\right) = -\frac{1}{w^2}\, dw.$$

Therefore dz has a pole of order 2 at ∞ and no zeros or poles elsewhere.

Proposition 6.1. *The divisor of dz is*

$$(dz) = -2[\infty].$$

Hence a canonical divisor on $\mathbb{CP}^1$ is

$$K \sim -2[\infty].$$

In particular,

$$\deg K = -2.$$

Proof. The computation in the coordinate w shows that near $w = 0$,

$$dz = -w^{-2} dw,$$

so $\operatorname{ord}_\infty(dz) = -2$. Elsewhere dz is a nowhere-vanishing holomorphic differential in the coordinate z. Hence $(dz) = -2[\infty]$. □

Remark 6.2. The identity $\deg K = -2$ is the genus-zero instance of the general formula

$$\deg K = 2g - 2.$$

7. Riemann–Roch on $\mathbb{CP}^1$

Theorem 7.1 (Riemann–Roch on the projective line)**.** *For every divisor D on $\mathbb{CP}^1$,*

$$\ell(D) - \ell(K - D) = \deg D + 1.$$

Proof. Let $d = \deg D$. Since $D \sim d[\infty]$ and $K \sim -2[\infty]$, we have

$$K - D \sim -(d+2)[\infty].$$

Therefore

$$\ell(D) = \ell(d[\infty]), \qquad \ell(K - D) = \ell(-(d+2)[\infty]).$$

We now consider three cases.

Case 1: $d \geq 0$. Then

$$\ell(D) = d + 1, \qquad \ell(K - D) = 0,$$

so

$$\ell(D) - \ell(K - D) = d + 1.$$

Case 2: $d = -1$. Then both spaces are zero, and again

$$0 - 0 = 0 = d + 1.$$

Case 3: $d \leq -2$. Then

$$\ell(D) = 0, \qquad \ell(K - D) = \ell((-d-2)[\infty]) = -d - 1,$$

whence

$$\ell(D) - \ell(K - D) = 0 - (-d - 1) = d + 1.$$

This proves the formula in all cases. □

Corollary 7.2. *For every divisor D on $\mathbb{CP}^1$,*

$$\ell(D) = \max\{\deg D + 1, 0\}.$$

Proof. This is the explicit dimension formula already established above. It may also be read off from Riemann–Roch because $\ell(K - D)$ vanishes precisely when $\deg D \geq -1$. □

7.1. Interpretation

The theorem on $\mathbb{CP}^1$ should be viewed as a complete and explicit model of the general theory.

- Meromorphic functions are rational functions.
- Divisor classes are controlled by a single integer, namely degree.
- The canonical divisor is $-2[\infty]$.
- The space $L(D)$ depends only on degree and can be written down by hand.

Thus the genus-zero case exhibits in its simplest form the central idea of Riemann–Roch: the dimension of a space of meromorphic functions is governed by degree, corrected by the canonical divisor.

8. A homogeneous viewpoint

It is often useful to describe sections of $\mathcal{O}(n)$ by homogeneous polynomials. If $n \geq 0$, every homogeneous polynomial

$$F(Z_0, Z_1) = \sum_{k=0}^{n} a_k Z_0^{n-k} Z_1^k$$

of degree n determines a section of $\mathcal{O}(n)$. On the chart U_0, dividing by Z_0^n yields the polynomial

$$F(1, z) = \sum_{k=0}^{n} a_k z^k.$$

Hence

$$H^0(\mathbb{CP}^1, \mathcal{O}(n)) \cong \{\text{homogeneous polynomials of degree } n \text{ in } Z_0, Z_1\}.$$

This identifies the complete linear system $|n[\infty]|$ with the linear system of degree-n homogeneous polynomials and leads naturally to the map

$$\mathbb{CP}^1 \longrightarrow \mathbb{P}^n, \qquad [Z_0 : Z_1] \longmapsto [Z_0^n : Z_0^{n-1} Z_1 : \cdots : Z_1^n],$$

whose image is the rational normal curve.

9. Fifty exercises

The following exercises are arranged from basic computations to more structural consequences. The first twenty are foundational, the next twenty develop fluency with divisors and line bundles, and the final ten emphasize synthesis.

Basic geometry and meromorphic functions

Exercise 1. Show that the two standard charts U_0 and U_∞ cover $\mathbb{CP}^1$, and verify directly that the transition relation on the overlap is $w = 1/z$.

Exercise 2. Identify the points $0 = [1 : 0]$ and $\infty = [0 : 1]$ in the two affine charts.

Exercise 3. Show that the map

$$\mathbb{C} \cup \{\infty\} \longrightarrow \mathbb{CP}^1, \qquad z \longmapsto [1 : z], \qquad \infty \longmapsto [0 : 1],$$

is a bijection.

Exercise 4. Compute $\operatorname{ord}_\infty(z)$, $\operatorname{ord}_\infty(1/z)$, and $\operatorname{ord}_\infty(z^m)$ for $m \in \mathbb{Z}$.

Exercise 5. Show that $\mathcal{M}(\mathbb{CP}^1) = \mathbb{C}(z)$ by proving that a meromorphic function on $\mathbb{C}$ with at most a pole at ∞ is rational.

Exercise 6. Let $P(z)$ be a nonzero polynomial of degree n with roots $a_1, \ldots, a_r$ of multiplicities $m_1, \ldots, m_r$. Prove that

$$(P) = \sum_{i=1}^{r} m_i [a_i] - n[\infty].$$

Exercise 7. Let $Q(z)$ be a nonzero polynomial of degree m. Compute the divisor of $1/Q(z)$.

Exercise 8. Show that

$$\left(\frac{z-a}{z-b}\right) = [a] - [b]$$

for distinct $a, b \in \mathbb{C}$.

Exercise 9. Let

$$f(z) = \frac{(z-a)^r (z-c)^s}{(z-b)^t}$$

with a, b, c distinct. Compute (f) explicitly.

Exercise 10. Verify directly from rational functions that every principal divisor on $\mathbb{CP}^1$ has degree zero.

Divisors and linear equivalence

Exercise 11. Show that every point $[a]$ with $a \in \mathbb{C}$ is linearly equivalent to $[\infty]$.

Exercise 12. Prove that every degree-zero divisor on $\mathbb{CP}^1$ is principal.

Exercise 13. Let

$$D = [a_1] + [a_2] + \cdots + [a_n] - n[\infty].$$

Construct explicitly a rational function f such that $(f) = D$.

Exercise 14. Given a divisor

$$D = \sum_i m_i [a_i] - \sum_j n_j [b_j] + c[\infty],$$

write down explicitly a rational function h for which $D - (h) = (\deg D)[\infty]$.

Exercise 15. Show that if $m[\infty] \sim n[\infty]$, then $m = n$.

Exercise 16. Deduce that the degree map defines an isomorphism

$$\mathrm{Pic}(\mathbb{CP}^1) \cong \mathbb{Z}.$$

Exercise 17. Let $D = 2[0] - [1] - [\infty]$. Show that D is principal and find one rational function with divisor D.

Exercise 18. Let $D = [0] + [1] - 2[\infty]$. Describe all nonzero functions $f \in \mathbb{C}(z)$ such that $(f) + 2[\infty] \geq 0$ and f vanishes at both 0 and 1.

Exercise 19. Suppose D and E are divisors on $\mathbb{CP}^1$ with $\deg D = \deg E$. Prove that $D \sim E$.

Exercise 20. Show that the Abelian group $\mathrm{Div}(\mathbb{CP}^1)/\mathrm{Prin}(\mathbb{CP}^1)$ is generated by the class of $[\infty]$.

The spaces $L(D)$

Exercise 21. Prove that if $D \sim E$, then multiplication by a suitable meromorphic function induces an isomorphism $L(D) \cong L(E)$.

Exercise 22. Show that

$$L(0) = \mathbb{C}.$$

Exercise 23. Show that if $n < 0$, then $L(n[\infty]) = 0$.

Exercise 24. Show that if $n \geq 0$, then $L(n[\infty])$ is the space of polynomials in z of degree at most n.

Exercise 25. Deduce that

$$\ell(n[\infty]) = \begin{cases} 0, & n < 0, \\ n+1, & n \geq 0. \end{cases}$$

Exercise 26. Let $D = [0] + [1] - [\infty]$. Compute $\ell(D)$ and produce an explicit basis for $L(D)$.

Exercise 27. Let $D = 2[0] + [1] - 3[\infty]$. Compute $\ell(D)$ and find a basis for $L(D)$.

Exercise 28. Let $D = [0] - 2[\infty]$. Determine all functions in $L(D)$.

Exercise 29. Show that if $\deg D < 0$, then $L(D) = 0$.

Exercise 30. Show that if $\deg D \geq 0$, then $\ell(D) = \deg D + 1$.

The bundles $\mathcal{O}(n)$

Exercise 31. Write out carefully the gluing condition for a holomorphic section of $\mathcal{O}(n)$ on the standard cover $\{U_0, U_\infty\}$.

Exercise 32. For $n \geq 0$, prove directly from the gluing condition that every holomorphic section of $\mathcal{O}(n)$ is represented on U_0 by a polynomial of degree at most n.

Exercise 33. For $n < 0$, prove directly from the gluing condition that $H^0(\mathbb{CP}^1, \mathcal{O}(n)) = 0$.

Exercise 34. Show that

$$h^0(\mathbb{CP}^1, \mathcal{O}(n)) = \begin{cases} 0, & n < 0, \\ n+1, & n \geq 0. \end{cases}$$

Exercise 35. Identify $H^0(\mathbb{CP}^1, \mathcal{O}(n))$ with the vector space of homogeneous polynomials of degree n in two variables.

Exercise 36. Show that every holomorphic line bundle on $\mathbb{CP}^1$ is isomorphic to $\mathcal{O}(n)$ for a unique integer n.

Exercise 37. Show that

$$\mathcal{O}(a) \otimes \mathcal{O}(b) \cong \mathcal{O}(a+b).$$

Exercise 38. Show that the dual bundle of $\mathcal{O}(n)$ is $\mathcal{O}(-n)$.

Exercise 39. Using divisors, explain why $\mathcal{O}(D) \cong \mathcal{O}(\deg D)$ on $\mathbb{CP}^1$.

Exercise 40. Show that the complete linear system $|n[\infty]|$ defines the map

$$\mathbb{CP}^1 \longrightarrow \mathbb{P}^n, \qquad [Z_0 : Z_1] \longmapsto [Z_0^n : Z_0^{n-1} Z_1 : \cdots : Z_1^n],$$

and describe its image.

Canonical divisor and Riemann–Roch

Exercise 41. Compute (dz) on $\mathbb{CP}^1$.

Exercise 42. Compute the divisor of $z^m\,dz$ for $m \in \mathbb{Z}$.

Exercise 43. Show that a canonical divisor on $\mathbb{CP}^1$ is linearly equivalent to $-2[\infty]$.

Exercise 44. Verify that $\deg K = -2$.

Exercise 45. Compute $\ell(K)$.

Exercise 46. Let $D = n[\infty]$. Compute $\ell(K - D)$ for all integers n.

Exercise 47. Verify Riemann–Roch directly for the divisors $D = -2[\infty]$, $D = -[\infty]$, $D = 0$, $D = [\infty]$, and $D = 2[\infty]$.

Exercise 48. Give a case-by-case proof of Riemann–Roch on $\mathbb{CP}^1$ using only the explicit formula for $\ell(n[\infty])$ and the identity $K \sim -2[\infty]$.

Exercise 49. Show that for every divisor D on $\mathbb{CP}^1$,

$$\ell(D) = \max\{\deg D + 1, 0\}.$$

Explain why this is special to genus zero.

Exercise 50. Write a short proof, in textbook style, explaining why $\mathbb{CP}^1$ is the natural first example for learning the Riemann–Roch theorem.

10. Hints to the exercises

The following hints are intentionally brief. They are meant to guide a first reading without replacing the proofs.

Exercises 1–10

1. Use the definitions $U_0 = \{Z_0 \neq 0\}$ and $U_\infty = \{Z_1 \neq 0\}$.
2. In the z-chart, $0 = [1 : 0]$ and in the w-chart, $\infty = [0 : 1]$ becomes $w = 0$.
3. Check injectivity and surjectivity separately.
4. Use $z = 1/w$ near ∞.
5. Study the singularity at ∞ via $w = 1/z$.
6. Factor the polynomial and examine $z = 1/w$.
7. Negate the divisor from the polynomial case.
8. Use the divisor of $z - a$ and subtract.
9. Add the contributions from zeros, poles, and ∞.
10. Write $f = P/Q$ with P, Q coprime.

Exercises 11–20

11. Use $(z-a) = [a] - [\infty]$.

12. First reduce the divisor to the form $\sum m_i[a_i] - n[\infty]$.

13. Take $f(z) = \prod_i (z - a_i)$ with multiplicities.

14. Use the proof of the theorem on linear equivalence.

15. Compare degrees.

16. Surjectivity comes from the theorem; injectivity comes from Exercise 15.

17. Try $f(z) = z^2/(z-1)$.

18. Such functions lie in $L(2[\infty])$ and must vanish at 0 and 1.

19. Reduce both divisors to $\deg D\,[\infty]$.

20. Use the preceding exercise and the definition of the quotient group.

Exercises 21–30

21. If $D - E = (h)$, multiply by h.

22. These are exactly the global holomorphic functions on $\mathbb{CP}^1$.

23. Any nonzero element would have to be holomorphic and vanish at ∞.

24. Pole order at ∞ controls the polynomial degree.

25. Count the basis $1, z, \ldots, z^n$.

26. Reduce to $[\infty]$ using linear equivalence.

27. Again reduce to $\deg D\,[\infty]$.

28. The condition at ∞ forces a polynomial degree bound, and the zero at 0 imposes divisibility by z.

29. Replace D by $(\deg D)[\infty]$.

30. Use Exercise 25 together with linear equivalence.

Exercises 31–40

31. Translate the transition function z^n into the relation between local representatives.

32. Write $s_\infty(w) = w^n s_0(1/w)$ and demand holomorphicity at $w = 0$.

33. The same relation forces all coefficients to vanish.

34. Combine Exercises 32 and 33.

35. Homogenize a polynomial in z.

36. Use $\mathrm{Pic}(\mathbb{CP}^1) \cong \mathbb{Z}$ and denote the degree-n class by $\mathcal{O}(n)$.

37. Tensor products multiply transition functions.

38. Dual bundles invert transition functions.

39. Use $D \sim (\deg D)[\infty]$.

40. The image is the rational normal curve.

Exercises 41–50

41. Write $z = 1/w$.

42. Combine $(z^m) = m[0] - m[\infty]$ with $(dz) = -2[\infty]$.

43. This is exactly the preceding proposition.

44. Read the degree from the divisor.

45. Since $K \sim -2[\infty]$, use the formula for $\ell(n[\infty])$.

46. Replace $K - D$ by $-(n+2)[\infty]$.

47. Compute both sides numerically in each case.

48. Split into the cases $n \geq 0$, $n = -1$, and $n \leq -2$.

49. The formula follows from $D \sim (\deg D)[\infty]$; the genus-zero feature is that degree alone determines the class.

50. Mention rational functions, divisor classes, the canonical divisor, and the explicit dimension count.

11. Concluding summary

On the projective line, Riemann–Roch is completely transparent because the geometry is controlled by a single point at infinity. The decisive facts are the following:

1. $\mathcal{M}(\mathbb{CP}^1) = \mathbb{C}(z)$.
2. Every divisor is linearly equivalent to $\deg D\,[\infty]$.
3. The spaces $L(n[\infty])$ are precisely the polynomial spaces of degree at most n.
4. A canonical divisor is $-2[\infty]$.

With these in hand, the formula

$$\ell(D) - \ell(K - D) = \deg D + 1$$

becomes an elementary computation. Precisely for this reason, the case of $\mathbb{CP}^1$ should be mastered before one approaches the theorem on higher-genus compact Riemann surfaces.